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## PROJECT AND TEAM INFORMATION

### Project Title

EduPlace Hub – AI Powered Learning & Placement Portal

### Student/Team Information

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|---|---|
| Team Name:<br>Team # (Mentor needs to assign) | StacktTitans<br>FS-VI-T114 |
| Team member 1 (Team Lead)<br>(Last Name, name: student ID: email, picture): | Raj , Ayushman ; 23021386 ;<br>ayushmanraj430@gmail.com<br> |
| Team member 2<br>(Last Name, name: student ID: email, picture): | Kadayan ,Gungun ; 230223654 ;<br>gungunkadyan01@gmail.com<br> |

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| Team member 3<br>(Last Name, name: student ID: email, picture): | Pandey , Arpita ; 23122539;<br>arpita.pan7662@gmail.com<br> |
| Team member 4<br>(Last Name, name: student ID: email, picture): | Rawat ,Prema ; 230221554 ;<br>230221554@geu.ac.in<br> |

# PROJECT PROGRESS DESCRIPTION (35 pts)

## Project Abstract (2 pts)

EduPlace Hub is a MERN stack-based full-stack web application designed to provide a centralized platform for subject learning and placement preparation. The system aims to simplify access to academic resources by organizing study materials in a structured and user-friendly manner.

It allows administrators to upload notes and topics dynamically, while students can access them in real time through an interactive interface. The application uses REST APIs for communication between frontend and backend and Supabase for efficient cloud-based data storage.

The platform also supports live updates for newly added content, ensuring users always have the latest information. Overall, the project demonstrates practical implementation of CRUD operations, API integration, and modern web technologies in a scalable and efficient system.

## Updated Project Approach and Architecture (2 pts)

EduPlace Hub follows a three-tier architecture consisting of frontend, backend, and database layers. The frontend is developed using React.js, providing an interactive interface for users. The backend is implemented using Node.js and Express.js, which manages API routing, business logic, and request handling.

Communication between frontend and backend is handled through REST APIs over HTTP. For data storage and management, Supabase is used as a cloud-based backend service, providing a PostgreSQL database along with built-in APIs. Supabase simplifies database operations such as storing, retrieving, and managing notes while ensuring scalability and security.

The system supports dynamic note creation and retrieval, allowing users to access updated content in near real time. An auto-refresh mechanism is implemented on the frontend to simulate live updates. Key technologies used include Express.js for backend development, Supabase for database and backend services, and CORS for secure communication.

## Tasks Completed (7 pts)

| Task Completed | Team Member |
|---|---|
| Designed and developed the frontend using React.js, including Home page, Admin dashboard, and user interface components. | Ayushman Raj , Arpita Pandey |
| Implemented dynamic UI features such as live updates and real-time display of newly added notes. | Ayushman Raj , Arpita Pandey |
| Developed backend using Node.js and Express.js with REST APIs for handling note creation and retrieval. | Gungun kadyan , Prema Rawat |
| Integrated Supabase database and established frontend–backend communication for storing and fetching data dynamically. | Gungun kadyan , Prema Rawat |

## Challenges/Roadblocks (7 pts)

- One major challenge was establishing proper communication between the frontend and backend. Initially, API requests were failing due to incorrect endpoints and CORS issues. This was resolved by properly configuring routes and enabling CORS in the backend.

- Another challenge was handling database connectivity and authentication errors while integrating Supabase. Issues like incorrect credentials and connection failures were fixed by verifying environment variables and database configurations.

- Implementing real-time updates was also difficult, as the system initially required manual refresh. This was addressed by adding an auto-refresh mechanism on the frontend, and future improvement includes using WebSockets for true real-time updates.

- Debugging API responses and identifying errors between frontend, backend, and database layers was time-consuming. This was solved using tools like Postman and browser developer tools for effective testing and debugging.

## Tasks Pending (7 pts)

| Task Pending | Team Member (to complete the task) |
|---|---|

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|---|---|
| Implement authentication system (login/signup) using secure methods such as JWT or Supabase Auth. | Ayushman Raj |
| Add Update and Delete functionality to complete full CRUD operations for notes management. | Arpita Pandey |
| Implement real-time updates using WebSockets (Socket.io) instead of auto-refresh for instant data updates. | Gungun |
| Deploy the application (frontend and backend) on cloud platforms and improve UI/UX responsiveness for mobile devices. | Prema Rawat |

## Project Outcome/Deliverables (2 pts)

The EduPlace Hub project results in a fully functional full-stack web application developed using modern technologies. The system provides a centralized platform for subject learning and placement preparation, enabling efficient access to academic resources.

Key deliverables include a responsive frontend interface built with React.js, a backend server developed using Node.js and Express.js, and a cloud-based database integration using Supabase for secure data storage and management. The application supports dynamic note creation and retrieval through REST APIs, demonstrating CRUD operations and seamless frontend–backend communication.

Additionally, the project includes an admin dashboard for content management, real-time update simulation, and a structured database design. Other deliverables include complete source code hosted on GitHub, detailed project documentation, and a presentation outlining system architecture, workflow, and implementation.

## Progress Overview (2 pts)

The EduPlace Hub project is currently in Phase 2, with most of the core functionalities successfully implemented. The frontend interface, including the Home page, Admin dashboard, and navigation system, has been completed. Backend development using Node.js and Express.js is also functional, with REST APIs supporting note creation and retrieval. Integration with Supabase for database management has been achieved, and frontend–backend communication is working properly.

However, some features are slightly behind schedule, such as implementing full CRUD operations (Update and Delete) and real-time updates using WebSockets. Currently, auto-refresh is used as a temporary solution.

## Codebase Information (2 pts)

- **Repository Link:** https://github.com/ayushmanraj25/EduPlace-Hub.git
- **Branch Used:** main (primary development branch)
- **Important Commits:**
    - Initial project setup with React frontend and Node.js backend
    - Implemented REST APIs for note creation and retrieval
    - Integrated Supabase database for cloud data storage
    - Added frontend–backend connectivity and dynamic data fetching
    - Implemented live updates feature on Home dashboard

## Testing and Validation Status (2 pts)

| Test Type | Status (Pass/Fail) | Notes |
|---|---|---|

| | | |
|---|---|---|
| API Testing (Postman) | Pass | GET and POST APIs tested successfully |
| Frontend UI Testing | Pass | UI rendering and navigation working properly |
| Database Integration Testing | Pass | Supabase connection and data storage |
| Error Handling Testing | Pass | Empty fields and API errors handled correctly |
| Real-Time Update Testing | In progress | Currently using auto-refresh, real-time not implemented |
| Update/Delete Functionality | In progress | Only Create and Read implemented, Update/Delete pending |
| Authentication Testing | In progress | Login/Signup feature not implemented yet |

## Deliverables Progress (2 pts)

The EduPlace Hub project has achieved significant progress in delivering its core components. The frontend interface, including the Home page, Admin dashboard, and navigation system, has been successfully developed using React.js. The backend server using Node.js and Express.js is functional, with REST APIs implemented for note creation and retrieval. Integration with Supabase has been completed, enabling efficient cloud-based data storage and management.

Dynamic data fetching and live update simulation have been implemented, allowing users to view newly added content without manual refresh. The GitHub repository has been maintained with regular commits, ensuring proper version control. Project documentation, including abstract, system architecture, workflow, and progress reports, has also been prepared.